Fine-Tuning a Geospatial Foundation Model for UK Crop Classification

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NC-UK Internship

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UKCEH

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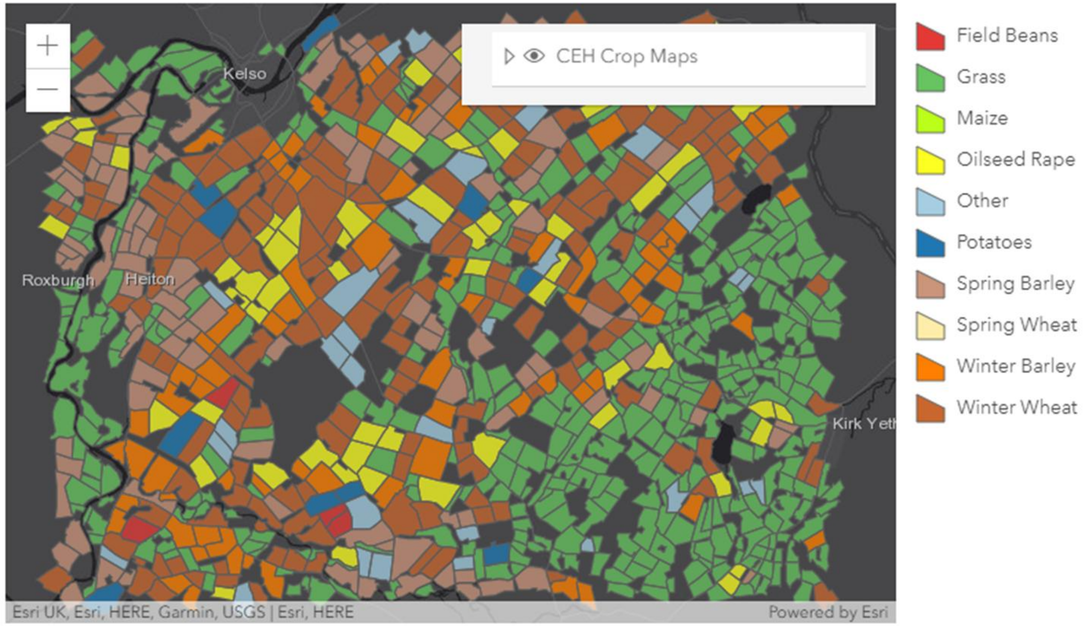
UKCEH Land Cover® Plus: Crops

- The UKCEH Land Cover® plus: Crops dataset is important for policy makers, regulatory agencies and scientists.
- Currently, UKCEH Land Cover® plus: Crops is updated using supervised learning methods that depend heavily on high-quality labelled training data. The existing method data consists of 1,655,942 fields covering an area of 9.7 million hectares.



https://www.ceh.ac.uk/data/ceh-land-cover-plus-crops-2015

Geospatial Foundation Models

Prithvi-EO-2.0: A Multi-Temporal Foundation Model for Earth Observation

- A vision foundation model trained on 4.2 million global time series samples from NASA's Harmonized Landsat and Sentinel-2 data archive.

- There are examples of Prithvi being used to effectively classify USA and other crops.

- The potential use and performance of Prithvi-EO-2.0 for UK crop classification was explored in this project.

Project Objectives

- Develop a proof-of-concept analysis demonstrating whether Prithvi EO-2.0 can be fine-tuned to perform classifications for a UK crop dataset.

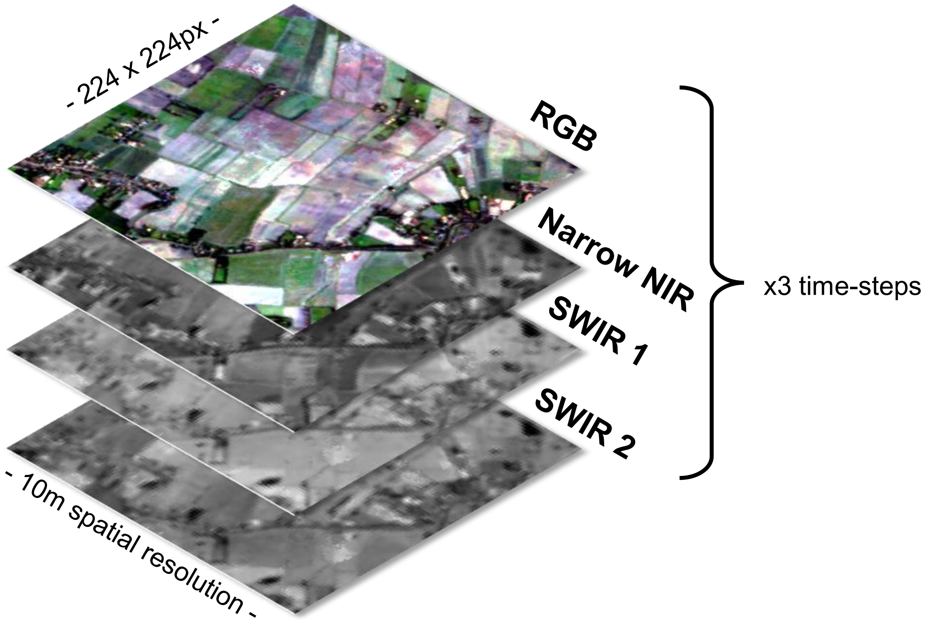
- Create a GitHub repository following UKCEH best-practice, containing clean, reproducible code using the TerraTorch package for fine-tuning Prithvi, plus saved configs and outputs.

Fine-Tuning Prithvi-2.0-EO

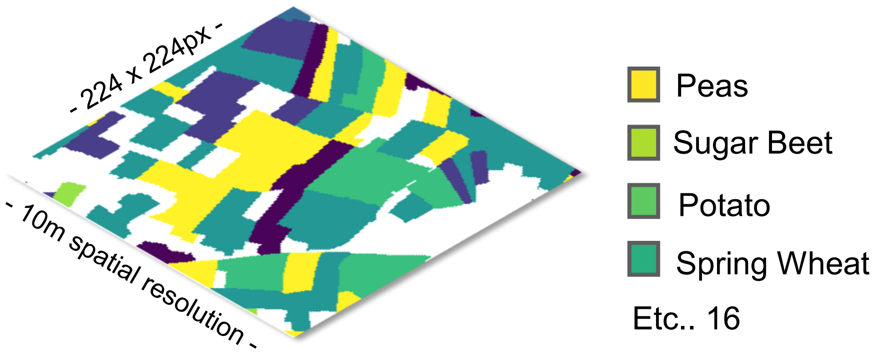
Dataset: Description

- ~900 raster GeoTIFF input chips (images and labels).
- Image chips were extracted from Sentinel-2.
- Label chips are from UKCEH Land Cover ® Plus: Crops classified into 16 classes.

Image



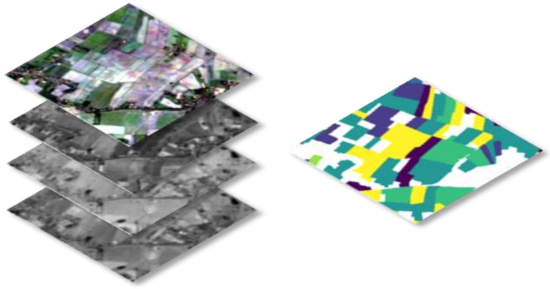
Label



Fine-Tuning Prithvi-2.0-EO

Fine-Tuning Pipeline

Fine-tuning a vision foundation model is the process of adapting a large, pre-trained computer vision model to excel at a specific downstream task.



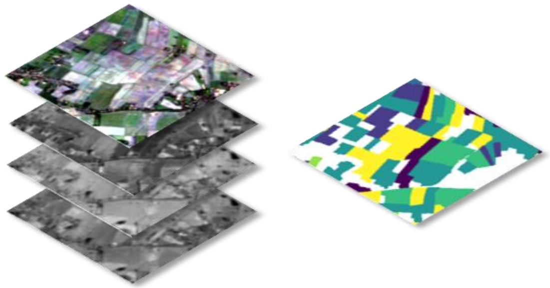
Prithvi-2.0-EO → Feature Embeddings → UNet Decoder → Prediction

- Training chips are passed through the pretrained Prithvi-2.0-EO frozen backbone that acts as a fixed feature extractor.
- Generates embeddings that capture the spatial, spectral, and temporal characteristics of the input imagery.
- Embeddings are passed to a trainable U-Net decoder, optimised to produce pixel-wise crop classifications.
- The final output is a semantic segmentation prediction in which each pixel is assigned to one of the predefined crop classes.

Fine-Tuning Prithvi-2.0-EO

Dataset Application: Training

- The ~900 chips were randomly split into **training** (60%), validation (20%), and testing (20%) subsets.
- Training was optimized with AdamW, at a learning rate of 0.0001 for 50 epochs, and a frozen backbone.
- Data augmentations such as random rotations were applied to minimise spatial biases.

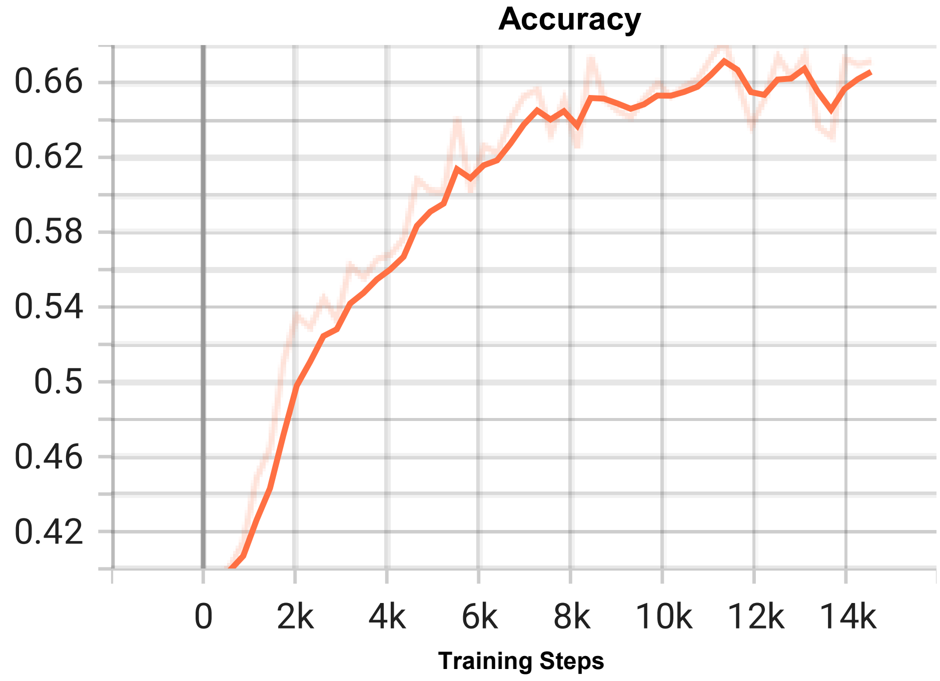


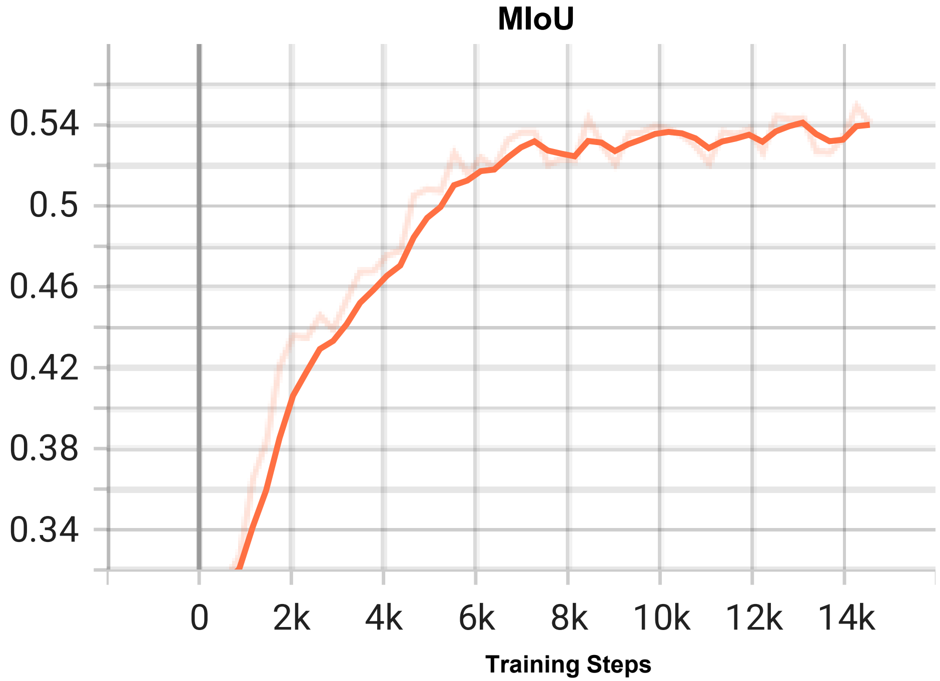
→ Prithvi-2.0-EO → Feature Embeddings → UNet Decoder → Prediction

Fine-Tuning Prithvi-2.0-EO

Validation Metrics

The ~900 chips were randomly split into training (60%), **validation** (20%), and testing (20%) subsets.





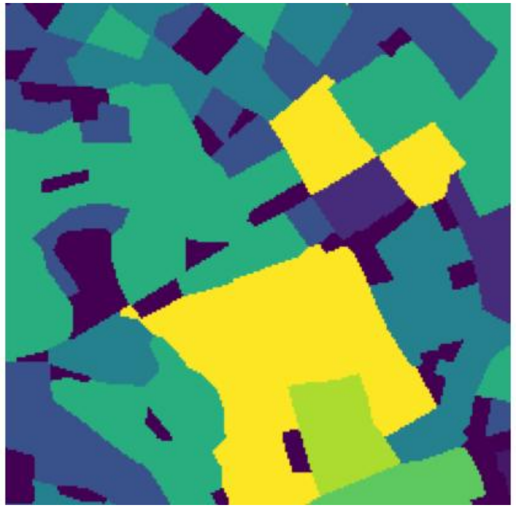
Fine-Tuning Prithvi-2.0-EO

Test Metrics & Visualisation

The ~900 chips were randomly split into training (60%), validation (20%), and **testing** (20%) subsets.

Prithvi-2.0-EO achieved **66.2% overall accuracy** and **53.9% mean IoU** across 16 classes, with a **macro F1 score of 66.9%.**

Test Image



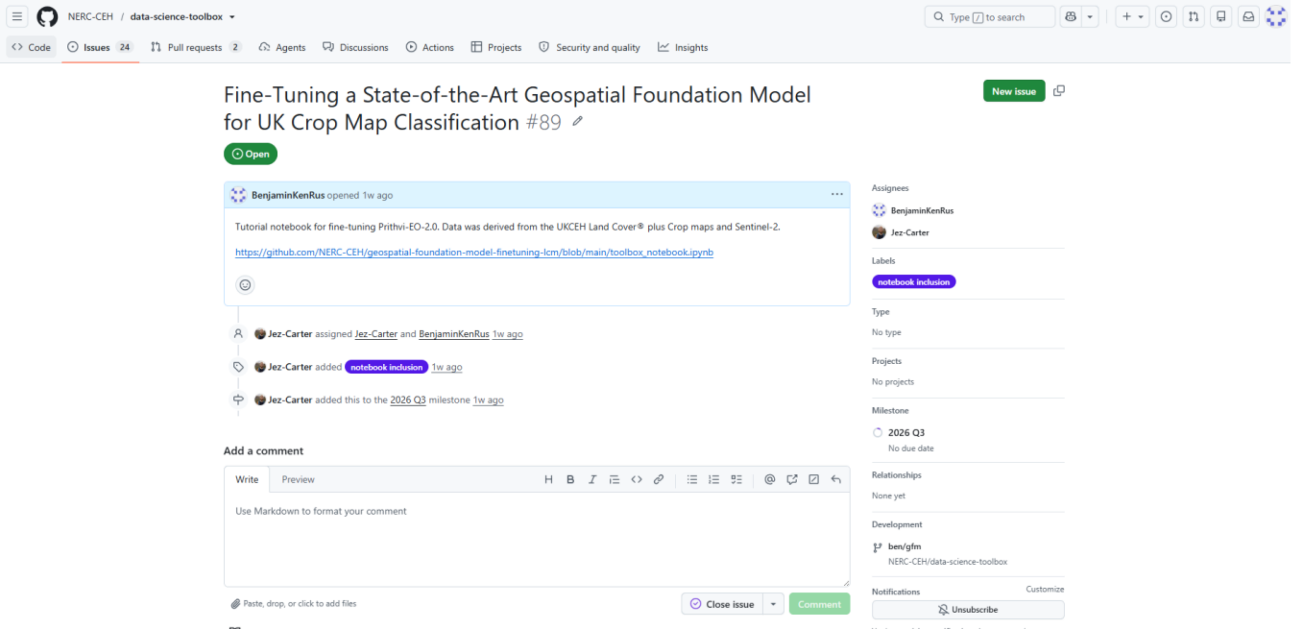
Ground Truth Crop Label

Model Prediction

Peas
Sugar Beet
Potato
Spring Wheat
Etc.. 16

Project Outcomes

This initial proof of concept project demonstrates that Prithvi-2.0-EO can learn useful representations for UK crop classification from a relatively small labelled dataset.



Project Outcomes

Areas of Improvement

- Larger dataset with more class balance, and less spatial leakage.
- Explore spatial resolution (compare 30m and 10m).
- Experiment with model parameters.
- Acquire validation data instead of pseudo data.
- More in-depth comparative analysis of existing Random Forest approach vs Prithvi-2.0-EO.

Thank You.

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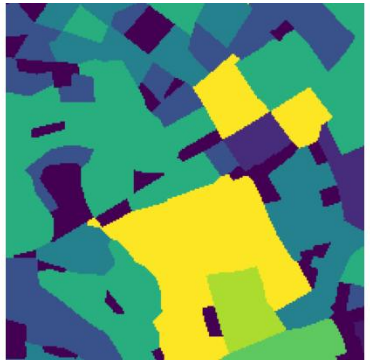
Fine-Tuning Prithvi-2.0-EO

Discussion

Prithvi-2.0-EO achieved **66.2% overall accuracy** and **53.9% mean IoU** across 16 classes, with a **macro F1 score of 66.9%.**

Test Image



Ground Truth Crop Label

Model Prediction